

Diversity Statement

Diversity is an inherently complex issue and everyone makes their own trade-off between ideal equality and dealing with historical inequality. It is exciting to see that Concordia University explicitly emphasizes the importance of campus diversity. Issues of diversity and inequality are deeply important to me - both personally and professionally. I want to briefly include my own take on this topic alongside what I hope to do. I believe that diversity is fundamentally the relationship between a community's distribution along a multitude of dimensions (e.g., physical traits, interests, cultures) and that of the general population.

Family Background. Being born and raised up in a small colony in a big city Kolkata, India I always saw people in my colony are helpful and having brotherhood relations with neighbors. I raised up between people having different religions and that made me a liberal person in nature. I grew up in a large and implausibly diverse family. My parents were progressive thinker who instilled in me the importance of community engagement. I have directly experienced economic uncertainty, which manifested in part as periodic lack of access to proper education and health care. My family's navigation of our life experiences reminded me daily of the ways that social inequities affect opportunities and security. My parents and my community inspired me to pursue what at times felt like a distant ambition of becoming an engineer who could address these inequities. I am a first generation to college individual from a lower middle class Indian family. It was very difficult for my family to provide college fund. Enduring energy, persistent ambition, and academic aptitude enabled me to complete my higher education. All my journey so far was a self-motivated journey of a small colony boy. My core values of social justice, service, and integrity have provided the compass that has guided me in my twenty-year academic journey. My commitment to promoting diversity and inclusion in academia and the engineering workforce is grounded in my personal experience, my many years of leadership and teaching, and my deep commitment to improving the computer, technology and statistical education of the diverse population of Mexico and the nation. As a faculty member at Centro de Investigación en Matemáticas (CIMAT), México and a consultant at Manufai, México, I have been fortunate to have the opportunity to do so not only as an individual, but also as part of a larger whole. I believe that engineering education has an essential role in cultivating an engineering workforce to meet the needs of our diverse communities.

Diverse Educational Background. I have built an academic career integrating my research and my wide-reaching public service, while engaging with communities, particularly communities of color and other priority populations. I can easily express my commitment to diversity, as I can also easily express my experiences of responding to the physical, emotional, social, and institutional barriers that constantly remind some of us of our differences. My life experiences have earned me an affiliation with a number of "marginalized groups". My commitment to diversity is informed by my experiences as an academican originated from India, a son of the parents migrated from East Pakistan, currently Bangladesh due to the partition of British-reigned greater India, having diverse educational background (Bachelor of Mechanical Engineering from Jadavpur University, India, Master of Technology in Quality, Reliability, and Operations Research, and Master of Technology in Computer Science from Indian Statistical Institute, as well as PhD in Computing Science from University of Alberta), multinational industrial work experiences (software industries like Robert Bosch India Limited, ITC Infotech India Limited, Cognizant Technology Solutions, Usha Martin Limited, Acellere, AQL Management Consulting Inc., Edmonton, Canada, and Manufai Mexico), teaching and research experiences from different universities (Canada, USA, Mexico and India), I bring additional insight to the diversity issues among diversified communities in Concordia University.

By publicly applauding individual eccentricities, I hope that it can encourage students to do the same. For example, weekly trivia posters about different community members may have a positive effect.

Multicultural Work Experience. I had the opportunity to work across different cultures and countries such as Canada, United States, Mexico, and India. The experience was enlightening as it enriched my perspective for research and advising. It also helped me appreciate the importance of diversity in science and engineering. Unfortunately, given the low representation of visually minor race in these fields, I believe mine was a very privileged experience. I am committed to achieving equity and enhancing diversity in science and engineering areas-ensuring that the representation of not only visually minor race, but also other minority groups is reflective of current times.

Teaching. As a graduate student, postdoc and faculty at different countries, I have had the opportunity to mentor several graduate students. As an representative of the visually minor group, the experience helped me to recognize the barriers faced by minor and other minorities engaging in computer science. As a faculty, I would be sensitive to these problems and create an environment of inclusion in my classes and research group where students from different background feel welcomed and can make progress. I will also actively recruit and train female and minority students in my area and help organize mentoring workshops to encourage them to pursue careers in the fields of artificial intelligence and industry 4.0. I would love to see non-computer science students such as artists, political science students, or architects take and excel in computer science classes. Computer science classes have historically been considered abstract, intellectual subjects, however big data and data science classes teach practical tools that are useful in any discipline. They could be the "gateway drug" we are looking for. For example, I am one of the founding members of the master program of computational statistics

(<http://mce.cimat.mx>) at CIMAT. A number of the students enrolled and finished were from non-computer science fields. One was even a non-technical CEO from a local business.

In my teaching, I take seriously the challenge of teaching to diverse audiences. As I explain in my teaching statement, I have found that my students have an enormous amount to teach me and I strive to learn from and to adapt both the style and content of my teaching material to reflect the diversity of my students. This has included mundane but important steps like learning to lecture more effectively to non-native English speakers as well as to create content and syllabuses that are more diverse in the material they include. In organizing classes, seminars, and lectures, I have found that more diverse programs are able to attract, and communicate more effectively to, broad audiences. By bringing in a wide variety of perspectives, a more diverse program can also support broader, more widely applicable, and more memorable learning.

I would also like to add new Diversity, Equity and Inclusion (DEI) courses. I would like to add classes that speak broadly about the topic and emphasize the tangible benefits of embracing diversity. Concepts such as growing the pie, and interest-based negotiation are tools that students can readily use, and highlight the value of embracing diversity from a practical perspective.

Creating Interactive Environment. I will strive to create a positive environment. This means I will highlight everyone's contributions, push shy or junior students to voice their opinions, speak up when there are subtle or blatant put-downs, support broad collaboration and encourage non-academic interests. These are minor details, but these everyday experiences do matter. For example, I started a chocolate fund that turned my office into a "watering hole" where students from every group on the floor comes to chat. Before this, cross-group interaction did not commonly occur. On a negative note, I have also seen students leave the graduate program because their group did not emphasize these qualities. My goal is for everyone passing through the university to accept that people vary widely along every attribute imaginable, and that this is valuable.

Advising. I mentor students on an ongoing basis; and on three separate occasions, I have received the very competitive basic science research fund to mentor and train individuals interested in pursuing research careers focused on discovering image based phenotypes that can substitute genomic biomarkers for making therapeutic and prognostic decisions of cancer and reduce the treatment cost. I recently mentored a student selected through competitive programa DELFIN whose commitment is to developing future leaders in the field of artificial intelligence. All of my mentees in these aforementioned programs are from groups underrepresented in science and research. My ongoing commitment to diversity, equity and inclusion is solid. Diversity, to me, is not just something to be achieved or talked about or theorized; it is the source of life for me.

In my 15 years in undergraduate and graduate engineering education I have directly observed the many ways that diverse teams make academia stronger. I have been fortunate to work alongside talented, creative and insightful trainees and faculty to solve complex problems. I have been inspired by the tenacity and honesty of our learners who overcame many obstacles to become engineering students, trainees and professionals, and who maintain deep commitments to social justice and their communities. As a supervising attending, I have learned both through direct observation and from the insightful academic work of my colleagues how cultural and language concordance improve the care we provide to our clients.

Since improving access of health care of diverse communities was my primary motivation for my research, it feels natural to dedicate myself to addressing equity and inclusion in all I do. As an advisory mentor for CIMAT, I had the honor of mentoring a number of students through significant personal and professional challenges and the good fortune of receiving exceptional peer faculty mentorship on how to do better for my advisees. I am committed to supporting our diverse trainees once they join our program and have gladly given my creativity, time and resources to that endeavor, one from which I have received great personal reward. I strive to maintain knowledge and sensitivity of the impact of race/ethnicity, gender, sexual orientation and gender identity on engineering education and in my daily teaching as well.

Volunteering. During my graduate study at University of Alberta, Each year I worked as a volunteer in the cross-cultural workshops organized by international student services. The goal of this program is to increase diversity and inclusion in AI development, policy, and research. At CIMAT, Mexico, as a part of the summer internship program, this year the plan is to host around 20–30 high school students, who belong to different racial minorities, in summer for 2–3 weeks. I plan to contribute to curriculum development for this program by adding a project that makes use of raspberry based smart-car to follow the line with obstacles, equipped with several sensors (infrared, ultrasound, and camera), that my students will set up with my supervision. Currently, the program has an acceptance rate of around ten percent. The low acceptance rate is attributable to the overhead of hosting students for the duration of this program. I believe to achieve the stated goal, improving the outreach of this program is critical. In the future, I plan to use my expertise in designing massive online courses, to develop an online version of this program to reach out to underrepresented student groups in remote locations.

Workforce Retraining. As the technology keeps evolving continuously, re-educating the workforce ensuring that they can catch up with its fast pace is a topic of national (and global) interest in current times. Big data discipline is an excellent example of how the nature of jobs evolves with time. Conventional data science jobs required a deep understanding of multiple complex data protocols. In contrast, many big data jobs now require expertise in programming languages, software engineering, parallel computing and relational data handling protocols instead.

In the past, I contributed to an online course on database management as well as advised students' master and doctorate thesis online. It provided me with an opportunity to reach out to tens of thousands of students from different age groups

and with diverse backgrounds. Many of them were mid-career data science employees honing their skills to embrace the changing ecosystem. Being able to contribute to such a massive workforce retraining was a gratifying experience for me. In the future, I plan to continue developing new courses that can contribute to such large-scale workforce retraining efforts.

Research. This commitment to diversity has also shaped my research. I take pride in the diversity of my group of coauthors. I grew up in a developing country where I experienced the unavailability of health care to the underrepresented community. In my medical imaging research, I aim to improve and reduce the cost of the image based diagnostic procedure for the treatment of different diseases such as cancer, inflammation, asthma and so on. I have dedicated my work in research and advocacy to framing the disproportionate burden of health care among marginalized communities as a social injustice. My research includes 1) potential association between image based phenotype and genomic biomarkers for making therapeutic and prognostic decisions of cancer, 2) quantification of white matter lesion between African American and White American, and 3) determination of pediatric brain age from Magnetic Resonance Imaging.

My current efforts continue to reflect my commitment to enhancing diversity, equity and inclusion. I have a long-standing commitment towards reducing inequities in engineering as well as improving equity and diversity in University faculty recruitment and retention. One of my main areas of research focuses on disparities in engineering and I have published numerous original research articles on that topic. One of these publications includes assessments of racial/ethnic differences in white matter lesion quantification.

Looking ahead to my new roles at the University, I intend to use my training, skills and position to ensure excellence and increase equity and diversity at my University and in society. I am very much interested to arrange such programs in future. I am an open-minded and liberal person in nature. I do not distinguish people by his/her sex, race, religion, political views or sexual orientation. I prefer to work in professional relationship but in a friendly way. I declare myself that I will strongly follow and respect the equity and diversity policy of the university.